pfi

# Python for Fixed Income

COURSE 1 · PYTHON FOUNDATIONS

# Setup: Python, VS Code, and Colab

WEEK 1 · DAY 00

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

# Two ways to take the course

| Route | What you need | Parts to follow |
|---|---|---|
| Colab, in your web browser | A Google account. Nothing is installed. | Part 5 only |
| Your own Windows computer | Python and VS Code, installed here | Parts 1 to 4 |

## Not sure? Start with Colab.

It is the quickest way in. Mac users: use Colab. You can set up your own computer later.

## On a work computer

Check your firm's rules before installing anything. Never upload work files to Colab.

Text too small or too big? Ctrl and + makes everything bigger, Ctrl and - smaller. In VS Code and every browser.

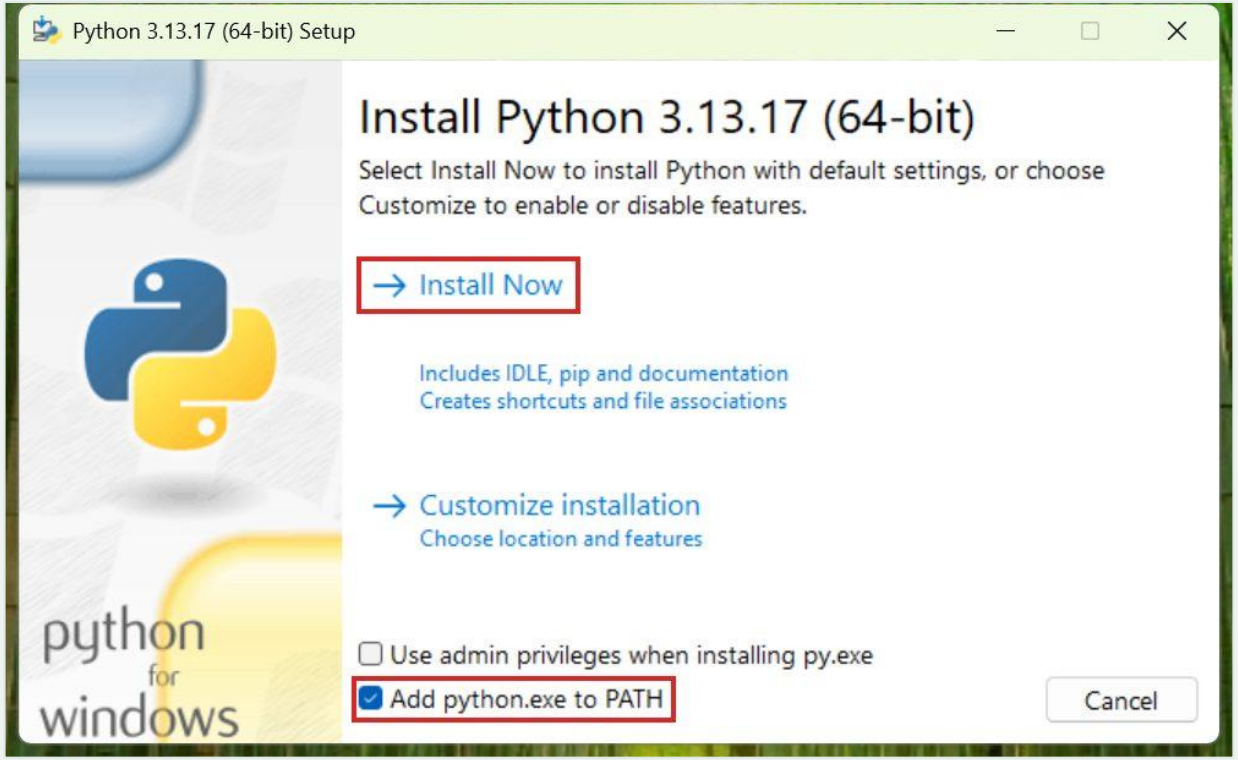
# Part 1: install Python

1. python.org/downloads/windows: under Stable Releases, the newest Python 3.13, Windows installer (64-bit).

2. Run it from Downloads.

3. Tick Add python.exe to PATH, at the bottom. Without it, Windows cannot find python or pip.

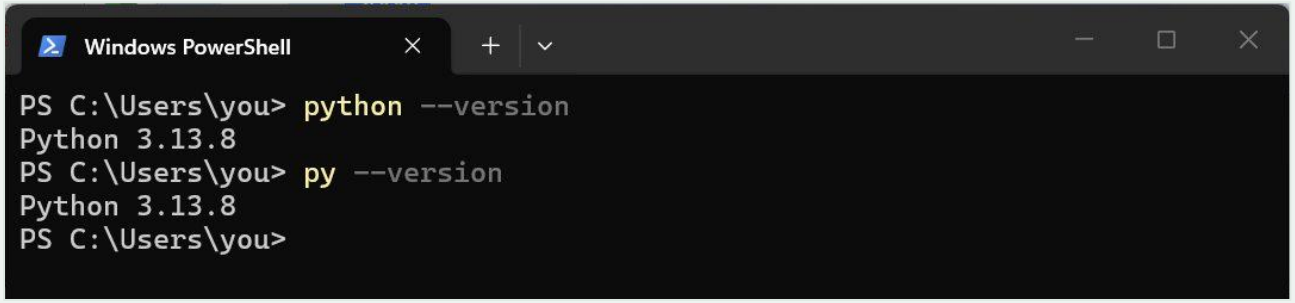
4. Click Install Now. Wait for Setup was successful, then Close.



# Check that Python works

Start, type PowerShell, open Windows PowerShell, then type:

```
python --version
```




| If you see | Do this |
|---|---|
| python is not recognized, or the Microsoft Store opens | Close PowerShell, open it again, retry |
| Still not found | Try py --version, and use py wherever the course says python |
| Neither works | Run the installer again with the PATH box ticked |

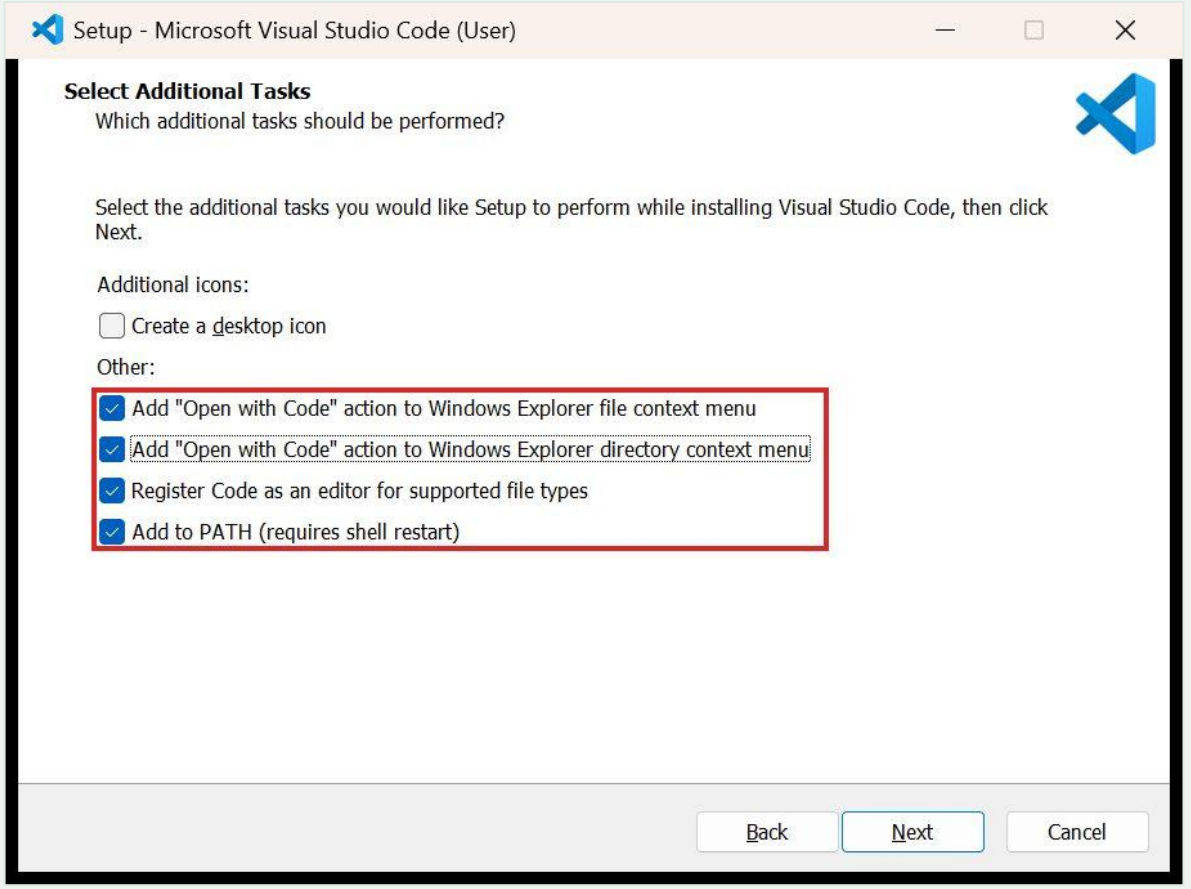
# Part 2: install VS Code

1. code.visualstudio.com/download: click Windows.

2. Run it. Accept the license, Next. Keep the location pages as they are.

3. Extra tasks: tick all four under Other, including Add to PATH.

4. Copilot page, if shown: leave it, Next. Then Install, and Finish.

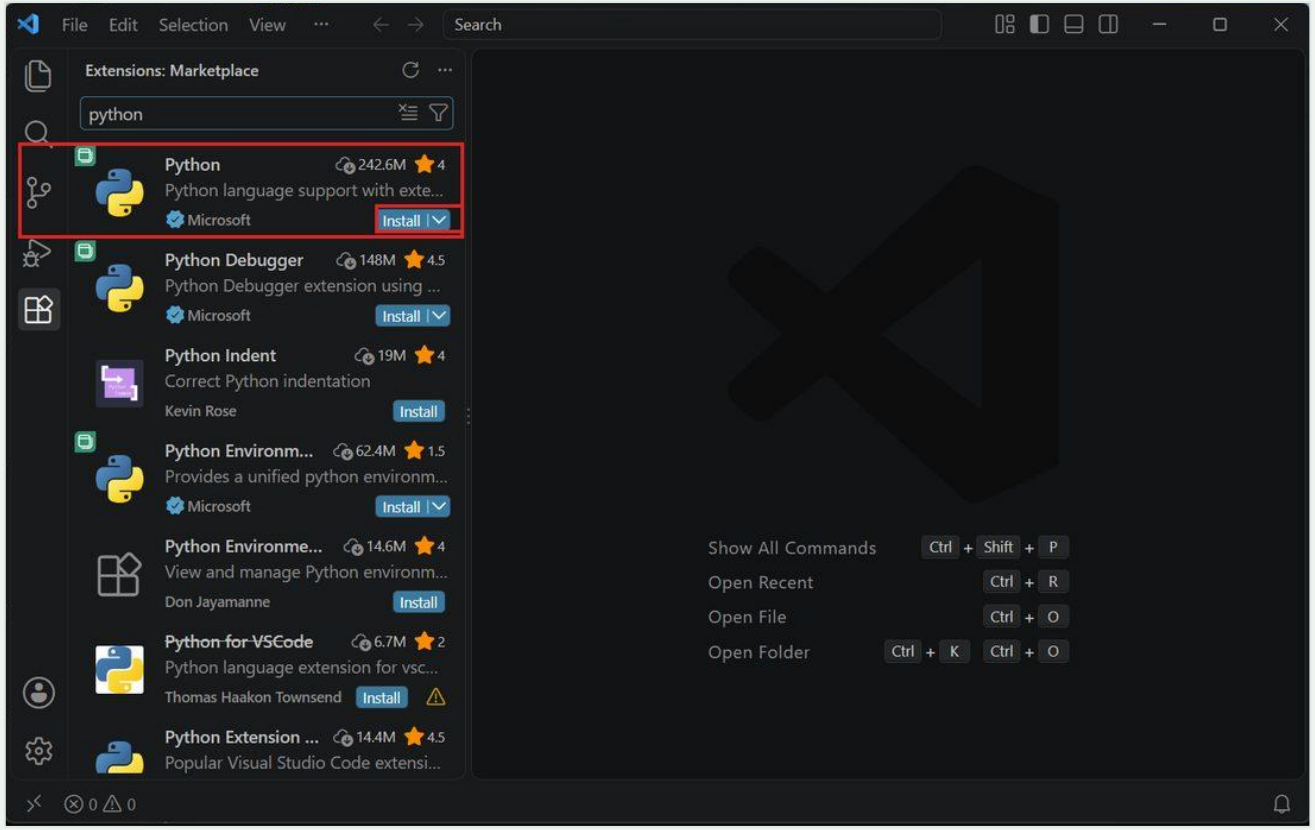


# Two extensions: Python and Jupyter

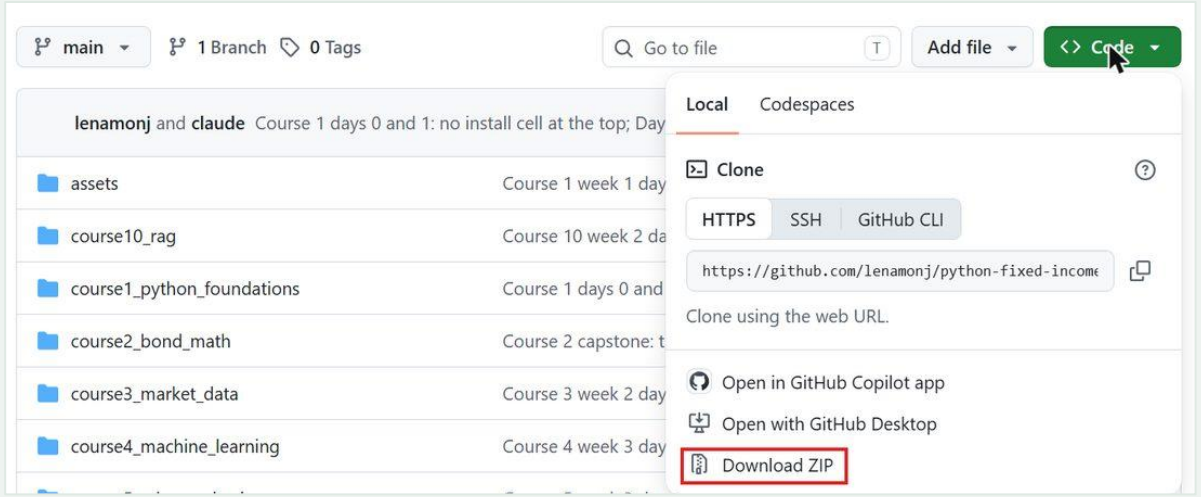
1. Click Extensions: the four small squares on the left (Ctrl + Shift + X).

2. Search python. Install Python, by Microsoft (blue tick).

3. Search jupyter. Install Jupyter, by Microsoft.



# Part 3: get the course onto your computer



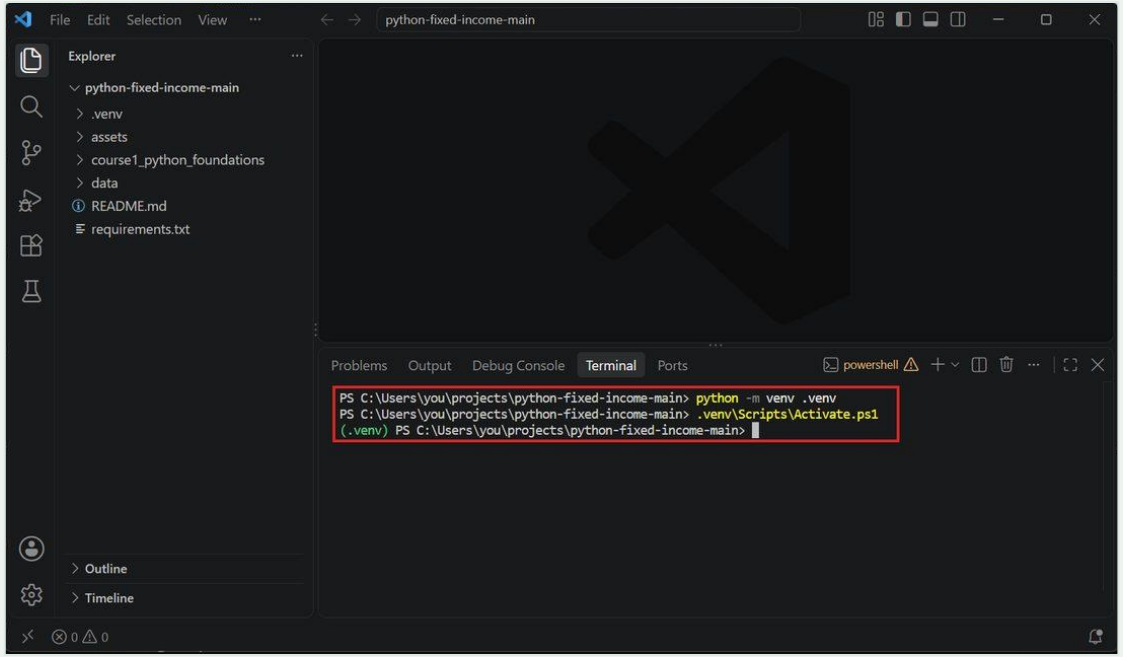

1. github.com/lenamonj/python-fixed-income: Code, then Download ZIP.
2. Make a folder called projects in your user folder.
3. Extract All the ZIP, and Browse to projects.
4. VS Code: File, Open Folder, python-fixed-income-main.

Check: the folder opens straight onto course1_python_foundations and data. A second folder of the same name inside means one level too deep.

# Part 4: the course environment

In VS Code: View, then Terminal. Type each line and press Enter.

```
Set-ExecutionPolicy -Scope CurrentUser RemoteSigned
python -m venv .venv
.venv\Scripts\Activate.ps1
python -m pip install -r course1_python_foundations\requirements.txt
```




Line 1 once per computer. The prompt then starts with (.venv). The install ends with Successfully installed.

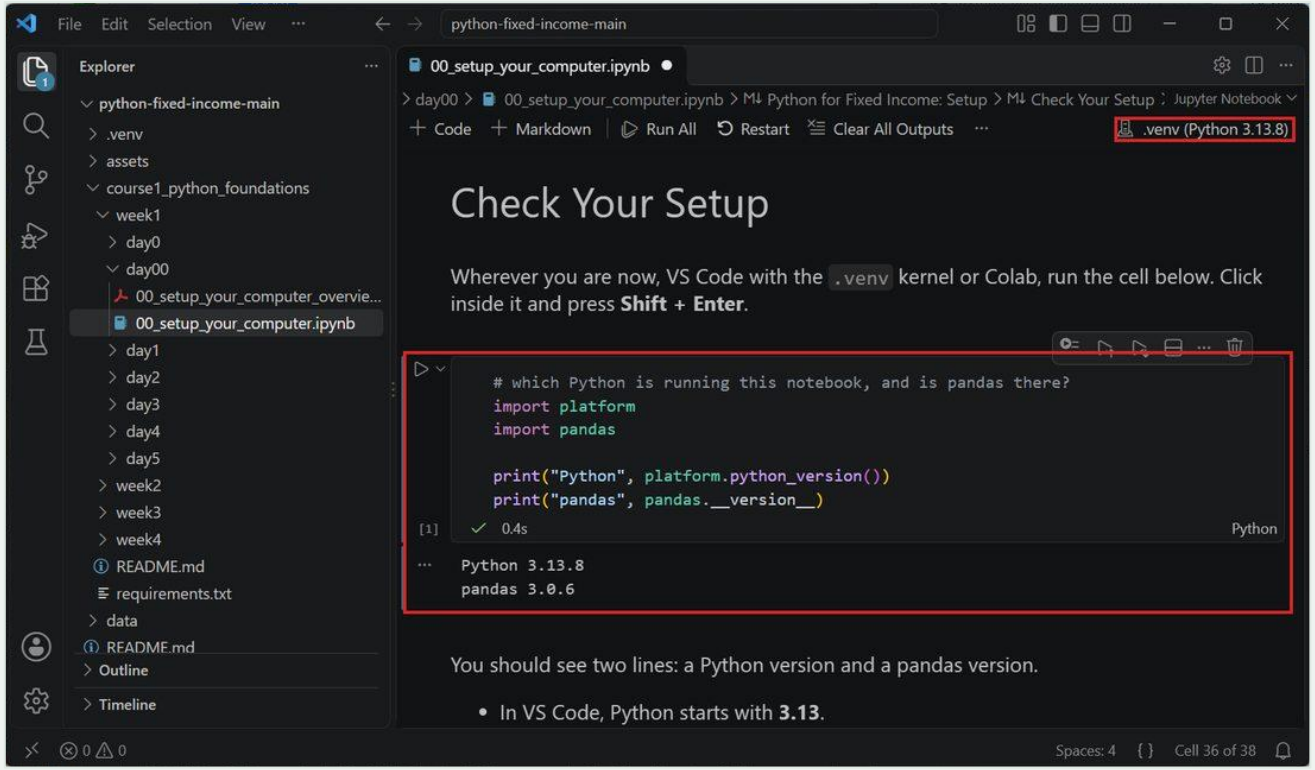
# Choose the kernel and run a cell

1. Open week1, day00, and this notebook, 00_setup_your_computer.ipynb.

2. Top right: Select Kernel, Python Environments, the .venv line (Recommended).

3. Scroll to Check Your Setup at the end. Click in its code cell, Shift + Enter.

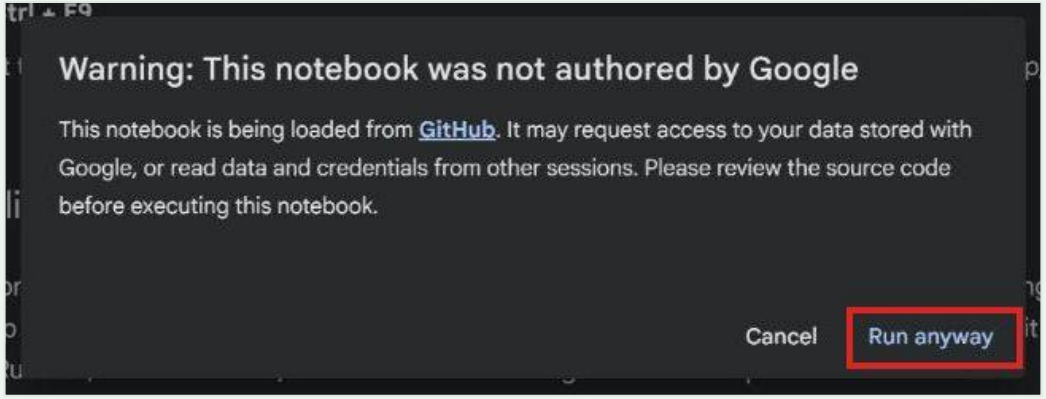
4. A Python 3.13 version and a pandas version appear. You are set up.




Shortcuts: Ctrl + and Ctrl - zoom. Ctrl + S saves. Shift + Enter runs a cell and moves on. Ctrl + Enter runs and stays. Ctrl + ~ opens the terminal. Ctrl + P opens any file by name.

# Part 5: Google Colab

| To | Do this |
|---|---|
| Open a course notebook | On GitHub, click its Open in Colab badge |
| Run a cell | Its round play button, or Shift + Enter |
| Run every cell | Runtime, then Run all (Ctrl + F9) |
| Keep your work | File, then Save a copy in Drive |
| Add a cell | Hover between two cells: + Code or + Text |




First run: this warning appears for every notebook from GitHub. Click Run anyway.

**Public or made-up data only in Colab, like the course's.**

# What you can do now

- Check Python with python --version
- Open the course folder in VS Code
- Switch on .venv and install the course's libraries
- Pick the .venv kernel and run a cell
- Open, run, and save a course notebook in Colab

**Day 0** Start Here: what Python, a notebook, and a library are